



Focus on innovation.

We take care of the rest.

`proto`, our AI engineer, plans every step.

`baker` makes the board. `mixer` tests it.

`protopus-kappa.vercel.app`

A prototyping lab and a full test bench, on one desk, run by an AI engineer

MORNING

You describe it

Proto reads the design, checks it and points at what will bite later.

MIDDAY

Baker builds it

Milled, masked, stencilled and pasted. You drop parts into a printed jig; Baker reflows.

AFTERNOON

Mixer tests it

One cable, eight instruments, and a verdict Proto writes in plain words.

EVENING

You try again

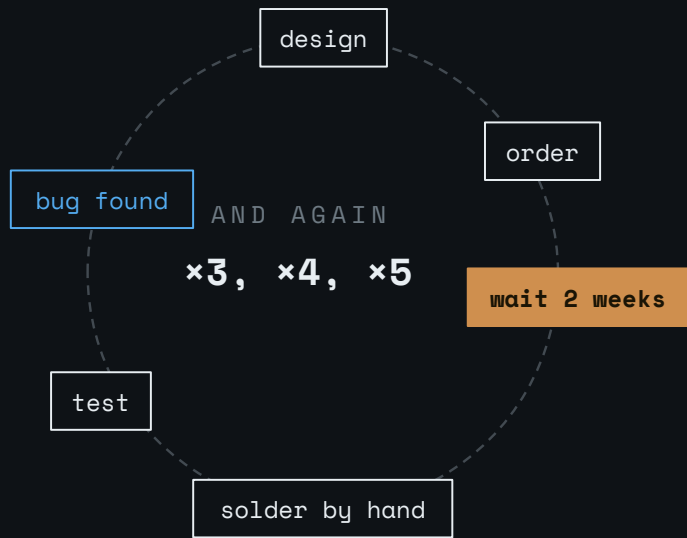
The next attempt starts the same evening, not two weeks from now.

WHAT MAKES IT OURS

Making and testing on one desk, with Proto guiding every step instead of a hundred buttons.

Built for prototypes: the first 1-10 boards of a design. Volume goes to a fab, with files that already worked.

A prototype is a loop, and every lap costs weeks



2

weeks waiting for a board to come back from a fabrication house

5

instruments to buy before you can even measure it

5

vendor apps and driver stacks to learn and keep alive

In my department a board ordered in week one gets measured in week four. **Nobody is short of ideas. They are short of tries.**

WHY NOW • compute is cheap enough for machines to carry the fidelity. People should carry the innovation.

AI can already build the software. Hardware is where it gets stuck.

SOFTWARE HAS A CLOSED LOOP

\$100M ARR

Lovable, 8 months after launch, selling to people who can't code. The model writes the code, runs it, reads the error and fixes it.

AI ALREADY DRAWS HARDWARE

Flux

\$49M raised · 1M+ users

Quilter

\$40M raised · AI layout

Diode

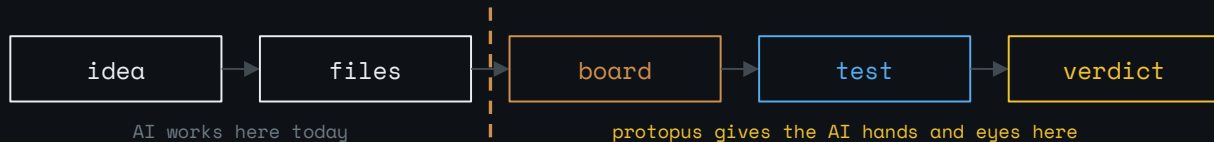
\$11.4M Series A · a16z

Claude ↔ Fusion, Blender

official connectors, 2026

...AND STOPS AT THE FILE

No model can make the board, power it, or see whether it works. The loop breaks at the bench, and a person carries it for two weeks.



Smart machines + smart accessories = the AI's access to the real world.

Every rung of hardware knowledge loses people. Proto removes rungs.

RUNG	YOU USED TO NEED	WHO CARRIES IT NOW
01 idea → spec	What exactly should it do?	Proto asks, then writes it
02 parts	Datasheets, stock, footprints	Proto picks in-stock parts
03 schematic	Circuit theory, references	Flux + Proto check it
04 layout	An EDA tool, DRC	Proto fixes it (Baker Studio)
05 fabrication	Vendors, customs, 7-20 days	Baker, same day
06 assembly	SMD soldering, stencil, reflow	Baker + printed jig
07 bring-up	Scope, logic analyser, firmware	Mixer measures
08 debug	Root cause, next revision	Proto explains the fix

Every rung we take away brings in a new crowd: students, designers, domain experts. Lovable showed what happens when the coding rung disappears.

HOW PEOPLE LEARN

By doing: try, look, adjust (Kolb's cycle). The speed of feedback sets the speed of learning.

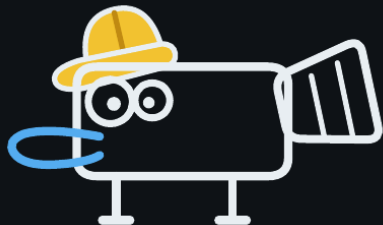
2 weeks

per try: beginners quit, experts batch changes

same evening

per try: a student gets many more attempts in one semester

One AI engineer, and the two smart machines it runs



proto

A platypus AI engineer inside both machines.

Reads the design · plans the job · flags mistakes · explains results



baker

Makes the board, ready for parts.

Mill · drill · mask · laser stencil · paste
· guided placement · reflow



mixer

Tests the board in real life.

Eight instruments behind one cable

VISION · a bridge between idea and innovation for anyone who wants to bring theirs to the world.

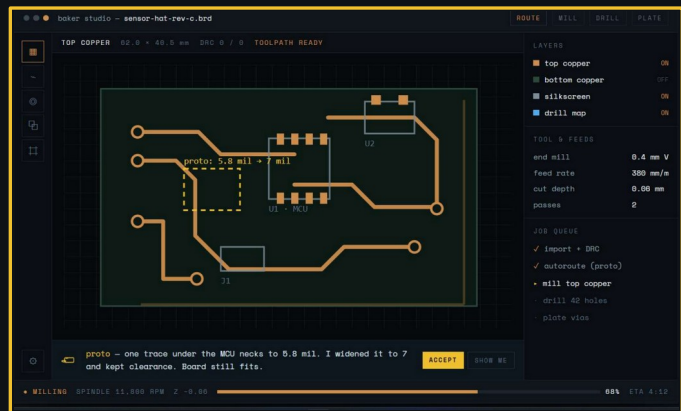
DEMO 1

Website walk-through ·
protopus-
kappa.vercel.app

DEMO 2

Video · raw Baker
prototype running

Proto: an AI engineer that understands your board



PROTOTYPUS Baker Studio

Proto has already found the thin trace, fixed it and planned the mill. You press Accept.

01

READS THE BOARD

Understands schematics and layouts. Pulls designs in through the Flux.ai API.

02

MAKES IT

Generates the Gerbers and the Baker job: mill, drill, mask, stencil, placement.

03

DIAGNOSES IT

Tests through Mixer, understands the waveforms and says in plain words what is wrong.

04

HAS A CHARACTER

A platypus engineer with a personality developed through Gemini.

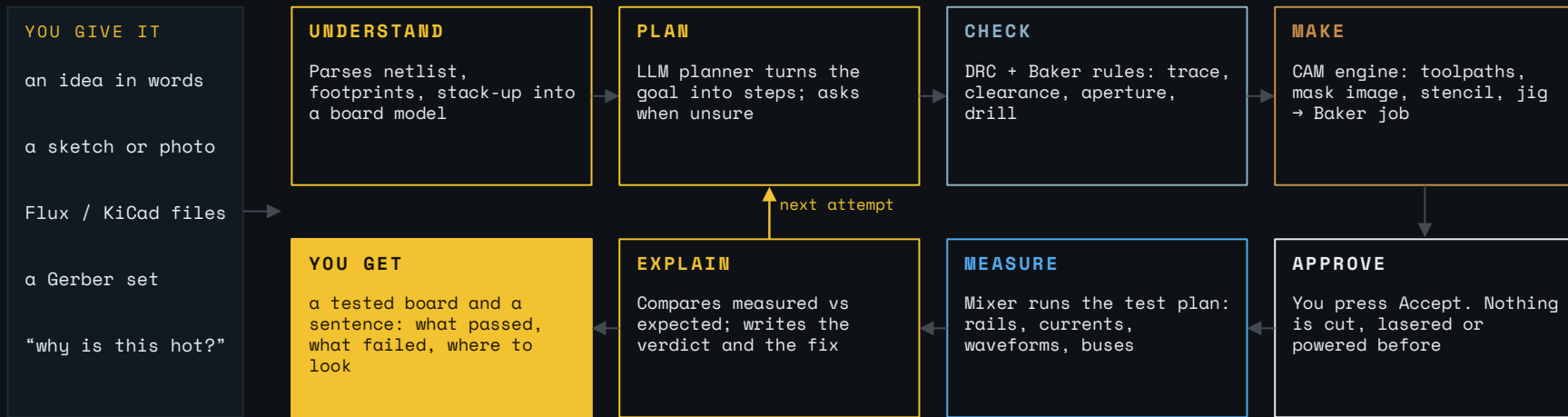
COMING NEXT

Probe matrix accessory

A grid of probes under the board gives Proto full access to every net on the PCB: it can measure anything, not just what you clip.

Simple enough that a **high-school student** can print a board.

Proto is a loop, not a chatbot: understand, plan, check, make, measure, explain



MEMORY

Every run is logged: design → process settings → measurements → outcome. The next board, and every other Baker, learns from it. No files-only AI tool can collect this data: it is the moat.

Model-agnostic: the character runs on Gemini today; the planner calls any frontier model through an API. Machine control stays local and works offline.

What happens between “import” and “PASS”

baker studio — sensor-hat-rev-c.brd · 62.0 × 40.5 mm

```

10:02  you      import sensor-hat-rev-c from Flux

10:02  proto     DRC 0/0. Baker rules: one trace under the MCU necks to 5.8 mil.

10:03  proto     widened it to 7 mil and kept clearance. Board still fits.  [Accept]

10:03  proto     job: isolation mill, 0.4 mm V, 2 passes · drill 42 holes · mask · stencil · jig

10:04  baker     steps 01-12 running · placement jig printing in parallel

11:40  you      drop parts into the jig pockets · vacuum pen

11:52  baker     reflow, Sn42Bi58 low-temperature profile

12:05  mixer     T2 power up: current limit set · idle current in range  ✓

12:06  mixer     T3 rails: 3V3 within limit  ✓

12:07  mixer     T6 I°C scan: no ACK from the sensor  ✗

12:08  proto     “SDA has no pull-up. R4 is lifted on one pad. Reheat R4 and retest.”

12:15  mixer     retest: sensor ACKs, reading plausible  ✓  PASS
  
```

WHAT PROTO CALLED

Flux.ai API

design in

DRC + Baker rules

found 5.8 mil

CAM engine

toolpaths, mask, stencil, jig

Klipper / Moonraker

ran Baker

Test-plan builder

from the netlist

Mixer driver

supplies, DMM, logic

Language model

the plain-words verdict

Illustrative run on the board shown in Baker Studio. Times are the engineering targets (~2 h Baker cycle, ~15-20 min hands-on); the proof of concept measures them.

Where the AI is, and where it isn't

THE MODEL DOES

- understands what you want
- reads the design and datasheets
- plans the job and asks questions
- drafts the test plan
- explains results and proposes the fix
- has a character (Gemini)

THE ENGINES DO

- DRC maths and Baker process rules
- toolpaths, G-code, mask and stencil images
- jig geometry and placement map
- test limits and pass/fail
- current limits on every supply
- interlocks: door, E-stop, laser cut-off

YOU DO

- describe the idea
- press Accept before anything physical
- drop parts into the jig
- decide on the fix
- own the design and the IP

The model never drives a motor directly. It proposes; tested engines compute; you approve; the machine executes.

V0 · TODAY

Baker Studio: DRC fix + mill planning, with Proto inside

V1 · PROOF OF CONCEPT

Full Baker job + Mixer test plan + plain-words verdict

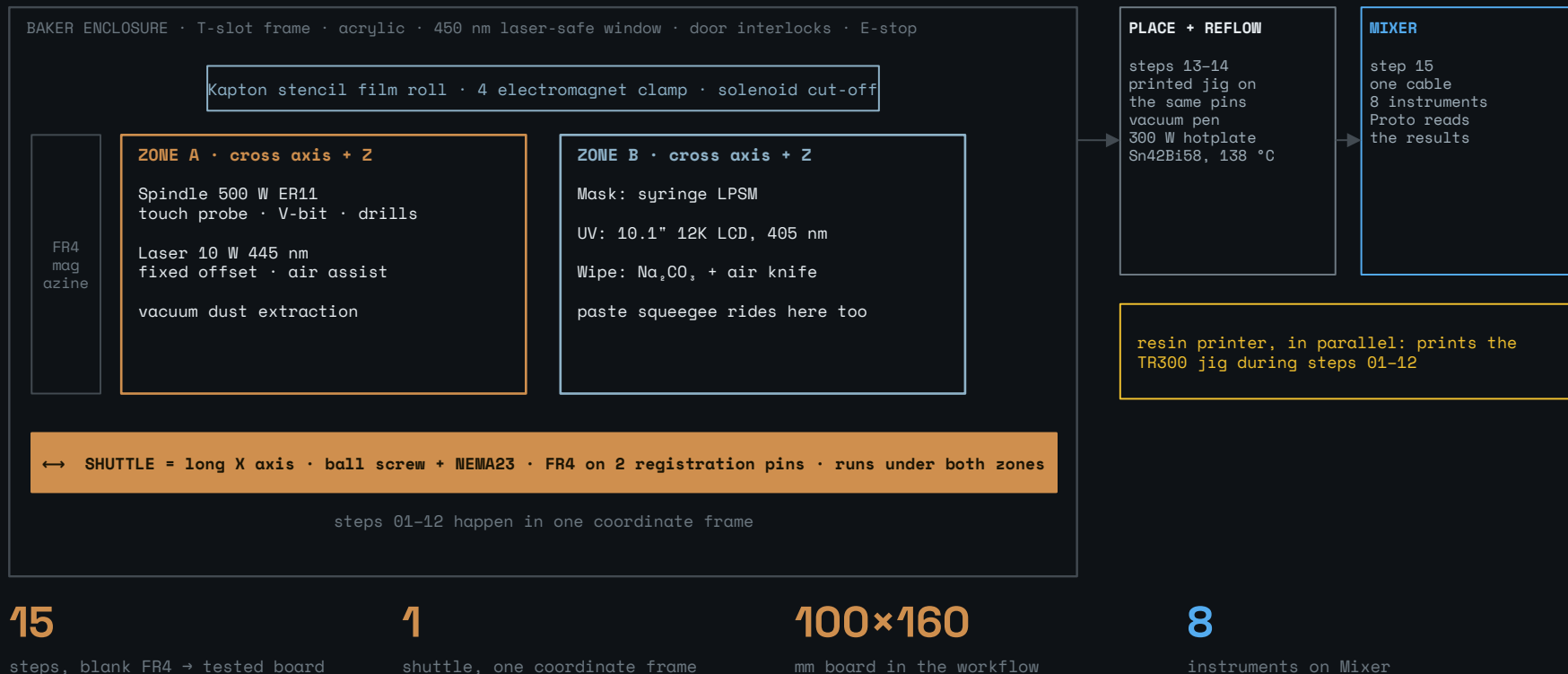
V2 · AFTER SEED

Idea → design via Flux/KiCad; enclosures via Fusion/Blender; 3D printer

V3 · FLEET

Learns from every Baker's run data; probe matrix sees every net

Fifteen steps on one shuttle, in one box



From a plain sheet to a board ready for Mixer

STEPS 01 · 05

It starts as a plain sheet

FR4 drops from the magazine onto two registration pins. One coordinate frame from here to placement.

cast · FR4 blanks · dowel pins · ball-screw shuttle + NEMA23

STEPS 10-12

The stencil fits this exact board

The laser cuts pad openings in film above the board, from the same Gerber in the same frame. Magnets clamp, squeegee pastes, film is cut away.

cast · 10 W 445 nm laser · Kapton roll · 4 × 8 kg magnets · 25 N solenoid

STEPS 02-04

Zone A reads the board, then cuts it

A touch probe maps height on a grid, so the 0.1 mm V-bit cuts the same few microns everywhere. Same spindle drills; vacuum pulls the dust.

cast · pogo probe · 500 W ER11 · 20° V-bits · 0.3-1.2 mm drills

STEPS 13

The jig does the aiming

While 01-12 ran, the resin printer made a jig with one pocket per part. You drop each part in with the foot-pedal vacuum pen.

cast · TR300 resin (160 °C) · vacuum pump + NC valve · foot pedal

STEPS 06-09

Zone B finishes the copper

Liquid mask, a 10.1" LCD cures all but the pads, a dilute Na_2CO_3 wipe lifts the pads clean, air knife, flood cure. No tank, no bath.

cast · syringe LPSM · 12K mono LCD · 405 nm UV · peristaltic pump

STEPS 14 → 15

Reflow at low temperature

The jig lifts off; the hotplate runs a Sn42Bi58 profile (138 °C), below the resin's limit, with less warp. Then Mixer tests it.

cast · 300 W silicone pad · SSR-25DA · MAX31855 + K-type

Same story, cheaper parts: ₹3,73,418 → ₹1,44,842

FUNCTION	PROTOTYPE (INDIA, 1 UNIT)	PRODUCTION (CHINA PARTS, 100)	WHY
Axes	Finished ball-screw linear stages	SFU1605 ball screws + HGR15 rails built in	≈₹5,500 per axis vs ₹16,500
Z axes	100 mm SFU1605 stage + NEMA17 ×2	T8 screw + linear guide	Spindle/laser Z and dispense Z
Control	Raspberry Pi 5 + BTT Octopus Pro + DM556	BTT Manta M8P V2 + CB1 (8 × TMC2209)	One board, 8 drivers, ≈₹10,800
Power	24 V + 48 V + 12 V supplies	One LRS-350-24 rail (spindle keeps its own)	Fewer supplies, simpler wiring
Spindle + laser	500 W ER11 kit · 10 W 445 nm diode (retail)	Volume-sourced; quotes at 100 units	Where the turnkey case is won or lost
Mask + UV	Syringe LPSM · 10.1" 12K LCD · ChiTu L K1 · 405 nm	Same engine	The LCD also prints jigs in production
Placement jig	Separate Saturn 3 Ultra + wash/cure	Small resin vat + lift under Baker's own UV screen	Removes the printer (to test)
Frame	T-slot extrusion + acrylic	Sheet-metal frame + acrylic panels	Repeatable fabrication
Firmware	Klipper + Moonraker + Proto job runner	Locked factory image; activation stays with us	IP by architecture
Unit cost	₹3,73,418 (incl. 35% duty/courier, 15% contingency)	₹1,44,842 (incl. duty, freight, Delhi assembly)	Deck target ₹1,50,000

Eight instruments, one cable, one brain



→ Proto compares every reading with what the design expects and writes the verdict in plain words

Analog Discovery 3 class. The prototype keeps 12/14-bit converters so the front end can be tuned first.

INSTRUMENT	PROTOTYPE SPEC
Oscilloscope	2 ch · 12-bit · 65 MS/s
Waveform generator	2 ch · 14-bit
Logic analyser	16 ch
Pattern generator	digital stimulus
Protocol decoder	I ² C · SPI · serial
Supply 1 + 2	set-point + current readback
Multimeter	24-bit

Status: architecture and BOM defined (*74,847 prototype); first build in the proof of concept.

T1 to T8, in the order a careful engineer would work

T1 Connect

One cable; probes on the test points the design calls for

T2 Power up safely

Supplies set; INA226 watches current, so a bridge is a spike, not a burnt part

T3 Check the rails

24-bit meter reads each rail and reference

T4 Drive the inputs

Generator + pattern generator push known signals

T5 Watch the outputs

Scope for analog, 16-ch logic for digital

T6 Read the buses

Decoder turns captures into I²C/SPI/serial traffic

T7 Verdict

Proto writes what passed, what failed, where to look

T8 Loop

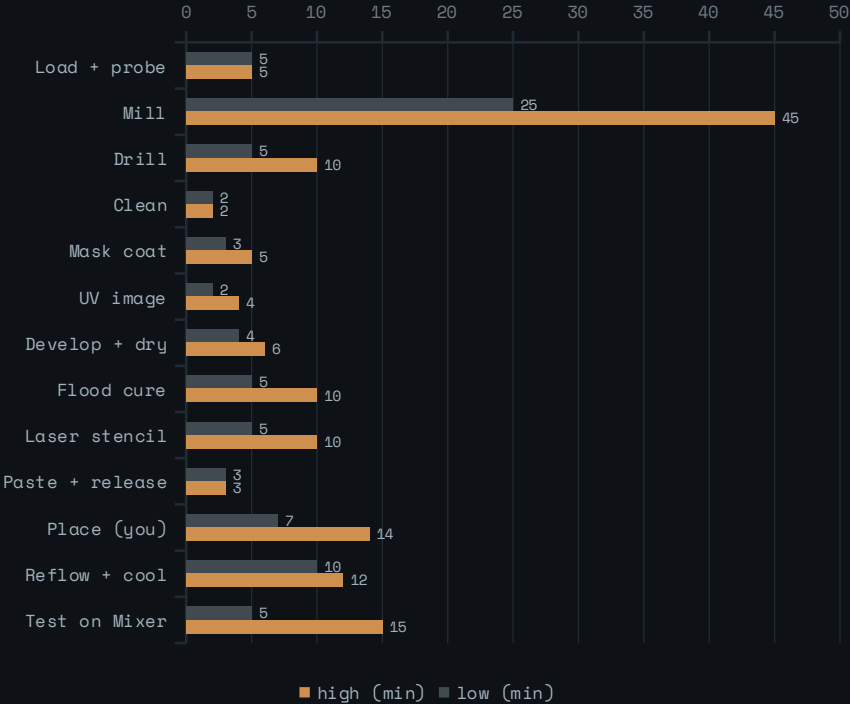
The fix goes into the design; next board the same evening

FUNCTION	PROTOTYPE	PRODUCTION	WHY
Capture	AD9226 12-bit 65 MS/s ×2	AD9288 dual 8-bit 80 MS/s	cost
Generation	AN9767 dual 14-bit	DAC908 8-bit 165 MS/s ×2	cost
FPGA	Tang Primer 20K Dock	Tang Nano 9K (GW1NR-9 + 64 Mb PSRAM)	smaller, cheaper
Host	Pi 5 + 7" screen	FT232HL USB 2.0 HS, no screen	one USB cable
Unit cost	¥74,847	¥13,740	target ¥15,000

ALSO OUR PRODUCTION TESTER

A golden Mixer + our test app tests every Mixer, and every Baker module, on the China line. The factory runs it; we own the limits, software and data.

About two hours on Baker; 15–20 minutes of it yours



81–141 min

per board, end to end

15–20 min

hands-on: load, place, hook up

	USUAL LOOP	BAKER + MIXER
Fabrication	outside fab, shipped	steps 01–12, on the bench
Stencil	ordered, aligned by hand	cut in place, same frame
Placement	tweezers or a service	printed jig + vacuum pen
Test	scope, supply, DMM, LA	Mixer: one cable
Diagnosis	engineer reads raw traces	Proto, in plain words
Next attempt	about two weeks later	the same evening

Engineering targets for a 100 × 160 mm board with ~40 parts; the jig prints in parallel. 3–5 boards per 8-hour shift. The proof of concept measures the real numbers.

Between the DIY bench and industrial manufacturing

	DIY bench	protopus	industrial
	breadboards, hand etching, borrowed scopes	proto + baker + mixer	fab house, assembly house, bench
TIME PER LOOP	1-3 days	Same day (~2 h on Baker)	1-3 weeks + customs
EXPERTISE	A seasoned engineer	Proto guides you	Expert to design and test
ASSEMBLED	By hand, if you can	Baker pastes + reflows; you drop parts into a jig	Yes, at extra cost
TESTED	Only if you own a bench	Yes, by Mixer	Separate ₹ lakhs of instruments
BEST FOR	one-off hacks	the first 1-10 boards of a design	100+ identical boards

The DIY bench needs an expert. The factory needs time and money. **Most ideas have neither.**

Others solve one step. Nobody closes the whole loop.

PRODUCT	PRICE · ₹	MAKE	ASSEMBLE	TEST	GUIDE
Voltera V-One	~3.1 lakh	●	●	○	○
LPKF ProtoMat	8.6–24.5 lakh	●	○	○	○
Makera Carvera	~5.1 lakh	●	○	○	○
Bantam Tools mill · discontinued	~3.1 lakh	●	○	○	○
Neoden YY1 pick-and-place	<2.6 lakh	○	●	○	○
Digilent Analog Discovery 3	~33,000	○	○	●	○
JLCPCB, PCBWay	per order, weeks	●	●	○	○
Flux (AI design)	1.3–4.3k / month	○	○	○	●
protopus proto + baker + mixer	3.34 lakh	●	●	●	●

List prices at ₹88 per US\$, before import duty. ASSEMBLE = paste + reflow or placement. Bantam Tools has stopped selling its PCB mill. Flux guides design and stops at the file.

voltera.io · Design World · Makera · ToolGuyd ·
EEVblog · Digilent · flux.ai

Not a desktop mill, not a factory. Where a fab wins, we say so.

BOARDS A YEAR → RIGHT TOOL

10	baker same-day loops beat fab minimums
100	baker one machine covers a busy lab
1,000	2 bakers + a fab any batch of 10+ identical boards → fab
10,000	fab / EMS production; Proto exports checked files
1,00,000	fab / EMS not what Baker is for

One Baker: 3–5 boards per shift, 750–1,250 a year. We scale by the number of labs, not boards per machine.

DESKTOP PCB MACHINES: A ROUGH HISTORY

Othermill → Bantam	PCB mill dropped; now sells art machines
Voltera V-One	out of stock; moved up to NOVA (\$45,974)
BotFactory	Squink \$3,199–4,499; SV2 now on request
Nano Dimension	sold its electronics unit, Apr 2026, \$2M cash

WHY, AND OUR ANSWER

one step automated → the whole loop, mill to verdict
fought \$2 boards on price → we sell hours per loop
a tool for experts → an outcome for non-experts
no software → Proto + run data are the moat

What we aim for: sell the machines, and one Proto plan runs all of them

PROJECTED PER UNIT · ₹	BAKER	MIXER
Hardware (target)	1,50,000	15,000
Assembly and QC	12,000	2,000
Packing and shipping	8,000	1,000
Warranty reserve	7,500	750
Software share	10,000	2,000
Sales and support	12,500	1,750
Projected total cost	2,00,000	22,500
Target selling price	2,99,000	34,999
Projected margin	~33%	~36%

Costed model: Baker ₹1,44,842, Mixer ₹13,740 incl. assembly.

ROADMAP TO THESE NUMBERS

NOW

Prototypes costed: Baker ₹3.73L, Mixer ₹74,847

MONTHS 0-9 · THIS ASK

₹16L proof of concept: Baker Rev B, 5 Mixers in studios

AFTER POC · SEED

~₹1.5 Cr: China pilots, first 100 Mixers + 20 Bakers

100-UNIT BATCHES

The projected unit economics above

PLANNED · ONE PROTO PLAN · EVERY MACHINE

₹9,999 / year

One subscription per workspace: Baker Studio, Mixer Station, Proto and every update.

Consumables

Board blanks, mask, paste, film, jig resin and end mills: ~₹334 of material per board, reordered as labs keep building.

Industrial variants, later

Bigger beds, multilayer and production test come after the base machines.

After about 53 boards, before counting the two weeks each loop saves

₹334

Baker consumables per board

₹6,721

the same assembled board from JLCPCB, landed in India

53

boards to recover ₹3,33,999 (Baker + Mixer)

COMPARED WITH	SAVING / BOARD	BREAK-EVEN	STARTUP · 2 / WEEK	LAB · 1 / WEEK
JLCPCB → India, 7-20 days	₹6,387	53 boards	6.6 months	13.3 months
Indian 24-hour fab, bare boards	₹5,166	65 boards	8.1 months	16.3 months

HOW ₹6,721 IS BUILT

5 PCBs \$5 · PCBA setup \$8 · stencil \$1.50 · 200 joints \$0.34 · 3 extended parts \$9 · DHL \$35 → ₹5,178 at ₹88, + 10% BCD and 18% IGST (₹1,543).

WHAT IT LEAVES OUT

Components (you buy them either way), engineer hours and the waiting. The Proto plan, ₹9,999 a year, is worth about two boards.

Start in the maker studios, then follow the people who watch them

₹6.4L cr global PCB market	₹3.1L cr global test and measurement	₹55K cr India PCB, ~15% a year	₹8.3K cr India test and measurement
33M+ active Arduino users worldwide	2.3M+ Hackster hardware community	10,000 Atal Tinkering Labs, 50,000 more planned	~2,500 Fab Labs in 125 countries

NOW

Maker studios

Free Baker and Mixer units, with a contract for featured videos. The Bambu Lab playbook.

NEXT

Hobbyists and startups

The people who watch those videos and lose weeks on every board.

ALONGSIDE

Prototyping labs

Tinkering labs, Fab Labs and college labs that already buy tools.

Where we are, told honestly

Baker

Raw prototype running on the bench. The video.

Software

Baker Studio with Proto inside. Website live.

Mixer

Idea phase. Architecture and BOM defined, not built.

SIGNAL

Met **Samuel Beek**, Founder & CEO of Schematik. Interested in funding us once the proof of concept is ready.

MONTH 0-2

Baker runs all 15 steps on a real board

MONTH 2-4

First Mixer built and measuring

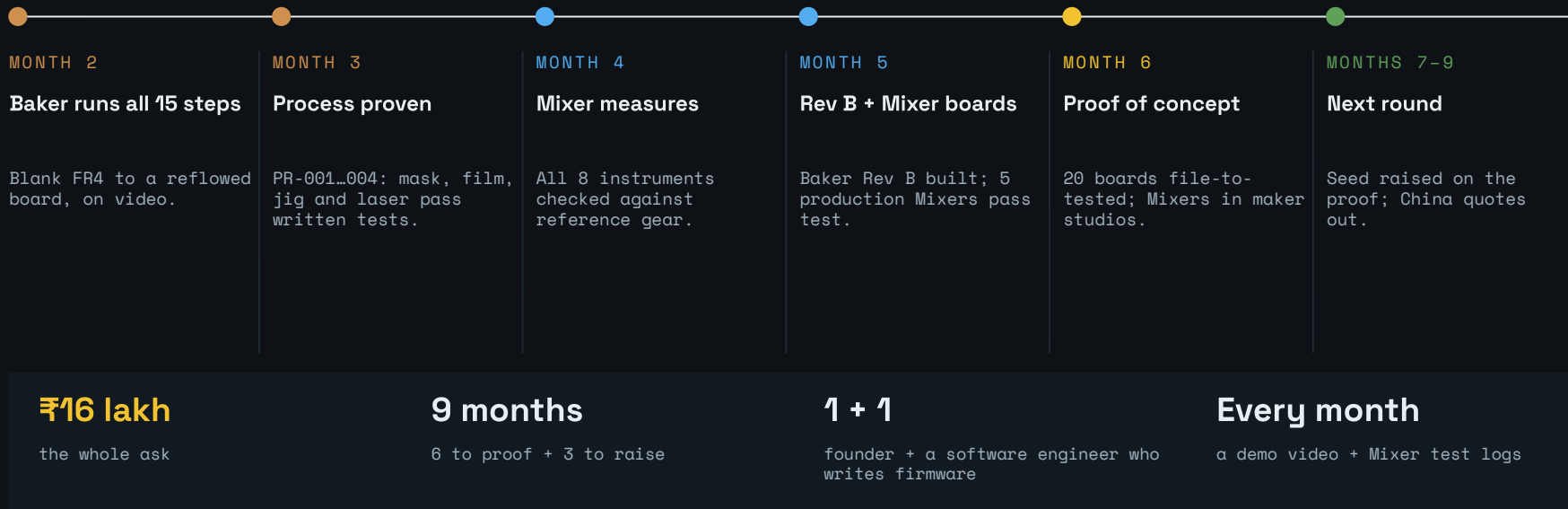
MONTH 4-6

Proof of concept, first maker-studio units

AFTER POC

Next round, first sales

Our goal: a working proof of concept, one milestone you can watch at a time



Built in the university lab, tested with students and makers.

Four open points, each written as a pass/fail test

ID	OPEN POINT	TEST ON THE PROTOTYPE	PASS	FALLBACK
PR-001	Wipe-develop cleanliness, 0.5 mm-pitch pads	0.5 mm footprint; inspect every pad after step 09	no mask left on any pad	skim pads with the spindle
PR-002	Film registration after magnetic clamping	measure film-to-copper offset at fiducials	±50 µm at every fiducial	clamp sequence / frame stiffness
PR-003	Jig-to-pin accuracy after resin shrink	post-cure jig; pockets vs pin datum	parts land on pads	scale compensation in the jig
PR-004	Laser offset drift with temperature	fiducial check, cold start and after a long job	inside the stencil budget	re-check every job (planned)

WHY THE ORDER IS THE DESIGN

mill before mask

the mask covers clean, finished copper

cure before stencil

the laser never touches uncured mask

stencil from the same Gerber, same frame

removes stencil-alignment error

jig prints in parallel

placement adds no waiting

Sn42Bi58 at 138 °C; jig lifts off

stays under TR300's 160 °C; less warp

Also to close before China: one board size, 100 × 160 (workflow) or 100 × 160 mm (BOM blanks). Plus lab safety: Class 4 laser inside an interlocked enclosure.

₹16,00,000

sixteen lakh: nine months to a proof of concept, and time to raise.

₹4.48L	₹2.20L	₹4.50L	₹1.35L	₹1.50L	₹1.97L
Baker + Mixer builds, costed line by line	Baker Rev B, 5 production Mixers, consumables	One software + firmware engineer, 9 months	Proto software and compute	Testing, tooling, lab safety	IP, outreach, 6.6% reserve
₹4,48,265	₹2,20,000	₹4,50,000	₹1,35,000	₹1,50,000	₹1,96,735

Enough to survive one failed iteration and raise from strength.

This builds the proof. The next round, at proof of concept, builds the business.

Where the ₹16 lakh goes, line by line

Baker		₹3,73,418	Mixer		₹74,847
Frame & enclosure	22,795		Capture + generation	18,198	
Motion	80,382		Front end + instruments	15,752	
Zone A: mill, drill, laser	38,270		Host, screen, I/O	26,752	
Zone B: mask, UV, wipe	33,058		Enclosure	1,510	
Stencil film	5,805		Parts	62,212	
Placement + reflow	4,891		Import duty + courier	2,872	
Board handling	300		Contingency (15%)	9,763	
Electronics & control	42,985		Prototype total	₹74,847	
Resin printer (jigs)	77,698		PRODUCTION UNIT (PROJECTED)		
First-run consumables	6,280		Baker		₹1,44,842
Parts	3,12,465		target ₹1,50,000 · under by ₹5,158		
Import duty + courier	12,247		Mixer		₹13,740
Contingency (15%)	48,707		target ₹15,000 · under by ₹1,260		
Prototype total	₹3,73,418				
			Use of funds		₹16,00,000
			Baker prototype build	3,73,418	
			Mixer prototype build	74,847	
			Baker Rev B iteration	1,00,000	
			Mixer production board ×5	60,000	
			PoC consumables	60,000	
			Engineer (software + firmware)	4,50,000	
			Proto software + compute	1,35,000	
			Testing + tooling	1,00,000	
			Lab set-up + safety	50,000	
			IP + legal	75,000	
			Demos + studio outreach	50,000	
			Reserve (6.6%)	71,735	
			Total	16,00,000	



People 28% · Hardware 28% · Iteration 14% · IP + reserve 12% · Lab + tooling 9% · Software 8%

Basis: PROTOPUS cost report, 29 Sep 2026: 96 BOM lines, 81 from live listings; USD at ₹88; imports carry 35% duty + courier; 15% contingency.

After proof of concept: raise, build in China, run lean from Delhi

THE ROUND

~₹1.5 crore seed,
at proof of concept

China pilots + first batches	₹70L
China set-up, IP	₹25L
Team of 5, 12 months	₹25L
Delhi centre + capital	₹20L
Buffer	₹10L

Seed round ₹1.5 Cr

Pilot batches are sellable (≈₹95L at list).
Schematik interested at PoC.

THE CHINA PLAN

Design in Delhi,
build in Shenzhen

- Mixer: full turnkey from batch one.
- Baker: China modules, final build in Delhi; turnkey once ≤ \$1,305 ex-works.
- 12 factories quoted → 2 pilots → 1 primary + backup.
- Pilots: 20/50/100 Mixers, 3/5/20 Bakers.

OPERATIONS

Delhi finalises,
it doesn't build

- Every unit: scan, test, activate Proto, ship.
- Mixer is our production tester: our rig, limits, data.
- Hire: a mechanical + a manufacturing/quality engineer.
- IEC, customs broker, China trademarks first.

MONTH 0-2

Design freeze, 12 RFQs out

MONTH 3-5

Pilots in China: EVT and DVT

MONTH 6-9

PVT: first 100 Mixers, 20 Bakers

MONTH 10-12

Mass production, first container

Months counted from the round. Round sizing is indicative; pilots costed from the BOM.

Two kinds of factory, one test system we own

MIXER LINE · FULL TURNKEY EMS, FROM BATCH ONE

M1 IQC	M2 SMT + AOI	M3 THT	M4 program	M5 PCBA FCT	M6 box build	M7 calibrate	M8 8-instr test	M9 serialise	M10 pack	M11 OQC	M12 ship FOB
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BAKER LINE · STAGE 1: TESTED MODULES (SKD) IN CHINA, INTEGRATION IN DELHI

B1 IQC	B2 frame	B3 motion	B4 zone A head	B5 zone B head	B6 stencil module	B7 control box	B8 module test (Mixer rig)	B9 kit + crate	D1 Delhi: integrate	D2 Delhi: calibrate	D3 Delhi: make- a-PCB
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INDIA FINALISATION CENTRE · NOT A SECOND FACTORY

I1 receive + scan	I2 quarantine	I3 power-on FCT	I4 calibration check	I5 activate Proto licence	I6 accept → stock
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WE OWN THE TEST SYSTEM

Golden Mixer + our test app: scan serial → run sequence → PASS/FAIL per step → upload. Operators can't edit limits. One record per serial (PTP-MXR-2027-000173): lots, firmware hash, calibration, every value.

IP BY ARCHITECTURE

The factory flashes a locked image; Proto activation and calibration logic stay with us. NNN in Chinese under PRC law, tooling is ours, no overruns, no substitution without a signed ECN.

Price the landed cost, not the factory quote

Baker

LINE	TODAY (REPORT)	CHINA TURNKEY
Parts	¥1,14,547 (\$1,302)	inside the ex-works price
Duty	¥14,775 (16.5%)	16.5% × EXW (HS to confirm)
Freight	¥3,520 LCL share	≈¥8,000 crated machine
India labour	¥12,000	≈¥3,000 acceptance
Landed	¥1,44,842	EXW × 1.165 + 11,000

Break-even EXW ≤ \$1,305 (≤ \$1,356 at the ¥1.5L target) vs \$1,302 of parts → start SKD; turnkey when quotes prove it.

Mixer

LINE	TODAY (REPORT)	CHINA TURNKEY
Parts	¥9,926 (\$112.8)	inside the ex-works price
Duty	¥1,638	16.5% × EXW
Freight	¥176	≈¥176 small carton
India labour	¥2,000	≈¥500 acceptance
Landed	¥13,740	EXW × 1.165 + 676

Break-even EXW ≤ \$127 (≤ \$140 at target): \$15-27 over parts. Tight at 100, comfortable at 1,000 → turnkey from day one.

NPI GATES · QUANTITIES SIZED TO EACH PRODUCT'S VALUE

GATE	MIXER	BAKER	QUESTION	EXIT (ALL MUST PASS)
EVT	10 × 2 suppliers	2-3 module sets each	Can they reproduce it?	meets test spec; Rev B drawings
DVT	50	5 machines	Right for production?	reliability, transport, EMC pre-scan; PR-001...004 on factory units
PVT	100, sellable	10-20, sellable	Is the process right?	line operators; yield ≥ target (e.g. 95%); signed authorisation

Source: China assembly line master plan #9 and #12. Freight and India acceptance are the plan's assumptions; duty rate from the cost report. About 12 months to the first container.



Thank you.

Come and try the loop.

protopus-kappa.vercel.app

soham9763015@ece.du.ac.in

MANAGEMENT TEAM & CORE SKILLS

Who is building this

Soham

Founder

Third year ECE, Faculty of Technology, University of Delhi

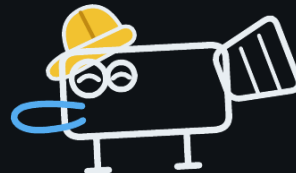
Industrial design

Mechanical architecture

Embedded electronics

Machine software

HIRING NEXT · a software engineer who also writes firmware.



APPENDIX · WHERE THE NUMBERS COME FROM

OUR DOCUMENTS	PROTOPUS Cost Report · BOM workbook · Baker + Mixer workflow (“The story of a PCB”) · China assembly line master plan · all 29 Sep 2026
FAB PRICING	jlpcb.com PCB and PCBA pages · zbotic.in India shipping, duty and rush-fab guides
MACHINES	voltera.io, eleshop.eu, elektor.com · Design World (LPKF) · Makera (PR Newswire, Jun 2026) · ToolGuyd, Make, Hackaday (Bantam) · TechCrunch (BotFactory) · Evertiq, 8 Apr 2026 (Nano Dimension) · EEVblog (Neoden)
INSTRUMENTS	Digilent AD3 (Jameco, DigiKey) · ADALM2000 (Mouser) · Moku:Go (TEquipment) · Red Pitaya · Saleae
AI	Flux (flux.ai; SiliconANGLE 27 Feb 2026) · Quilter (BusinessWire 7 Oct 2025) · Diode (Fenwick) · claude.com/connectors/autodesk-fusion · digitalproduction.com 30 Apr 2026 (Blender) · Lovable (TechCrunch 23 Jul 2025; founded.com)
MARKET	IMARC · Grand View · Arduino · Hackster · Atal Innovation Mission · Forbes · Fab Foundation · AICTE
CONVENTIONS	USD at ₹88 throughout, to match the cost report. Cycle times are engineering targets until the proof of concept measures them.